AUTOMATED LASER + MOLTEN SALT EXTRACTION PROCESS FOR WASTE PCB

1. COLLECTION OF WASTE PCB



Collection of discarded PCBs from various sources.

2. INITIAL INSPECTION & SORTING



Visual inspection and sorting to remove non-PCB materials and contaminants.

3. CLEANING & SURFACE PREPARATION



Removal of dust, dirt, solder flux and organic contaminants.

4. AUTOMATED LASER SCANNING & COMPONENT DETECTION



High-resolution laser scanning identifies and maps all components and target areas.

5. PRECISION LASER REMOVAL OF ELECTRONIC COMPONENTS



Automated laser precisely removes components (ICs, capacitors, resistors, connectors, etc.). without damaging the board.

6. SEPARATION OF METALLIC AND NON-METALLIC FRACTIONS



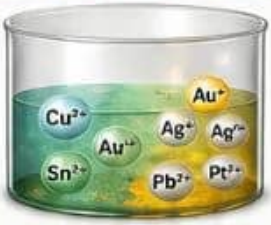
Mechanical processing (shredding, crushing & sieving) to separate metal-rich fraction from non-metallic materials.

7. MOLTEN SALT REACTOR (HIGH TEMPERATURE)



Metal-rich fraction is introduced into molten salt bath at high temperature (400–800 °C).

8. SELECTIVE DISSOLUTION OF TARGET METALS



Target metals dissolve selectively in molten salt forming metal ions/complexes.

9. METAL RECOVERY & SEPARATION PROCESS

BASE METALS (Cu, Ni, Sn)



Electrowinning / cementation to recover base metals.

PRECIOUS METALS (Au, Ag, Pd, Pt)



Chemical precipitation / solvent extraction / reduction to recover precious metals.

OTHER METALS (Pb, Zn, etc.)



Precipitation or solvent extraction to recover other valuable metals.

RESIDUES



Inert residues and slag are treated and disposed in an environmentally safe manner.

10. PURIFICATION (ELECTROREFINING / CASTING)



Purification of recovered metals to achieve high purity. Casting into ingots or pellets.

11. HIGH-PURITY RECOVERED METALS



Copper (Cu) · Gold (Au) · Silver (Ag) · Palladium (Pd) · Platinum (Pt) · Nickel (Ni) · Tin (Sn)

12. REUSE IN ELECTRONICS MANUFACTURING



Recovered metals are reused in the manufacturing of new electronic components and devices.

METALS RECOVERED

- Copper (Cu)
- Gold (Au)
- Silver (Ag)
- Palladium (Pd)
- Platinum (Pt)
- Nickel (Ni)
- Tin (Sn)
- Lead (Pb)
- Zinc (Zn)
- Others (Ga, In, Ta, Nb, etc.)



KEY FEATURES

- ✓ High recovery efficiency
- ✓ Selective metal extraction
- ✓ Reduced environmental impact
- ✓ Low energy consumption (laser precision)
- ✓ Suitable for complex PCBs

Conserves Natural Resources · Reduces E-Waste & Pollution · High Value Recovery · Sustainable & Environment Friendly · Cost-Effective Process